

UNIT-5

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DATE:

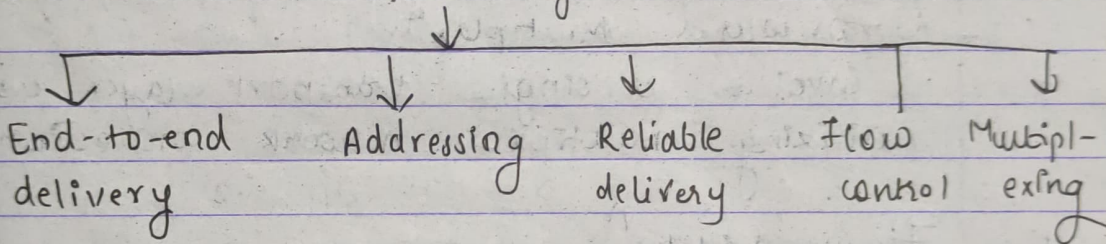
Q Explain services provided to upper layer by Transport layer.

- ⇒
- The transport layer is 4th layer from top.
 - main role is to provide communication services directly to application processes running on different hosts.

Services:-

services provided by Transport layer is similar to data link layer.

Transport layer services.



1) E-E-E d^y:-

The transport layer transmits entire message to the destination. Therefore, it ensures end-to-end delivery of an entire message.

2) Reliable:-

provides reliability services by retransmitting the lost and damaged packets.

It has 4 aspects:-

1) Error control

2) Sequence control

3) Loss control

4) Duplication control.

* Multiplexing :-

△ T.L uses multiplex^g to improve transmission efficiency.
It has 2

- Upward Multiplex^g :-

In .., multiple transport layers connections share same network connection. By utilizing upward multiplex^g, T.L can send multiple transmissions bound for same destination along same path.

- Downward Multiplex^g :-

involves a single transport layer connection utilizing multiple network connections.

* Flow Control :-

used to prevent sender from overwhelming the receiver. If receiver is overloaded with too much data, then receiver discards the packets & asking for retransmission of packets. This involves network congestion & thus, reducing system performance.

* Addressing :-

When data generated by an application on one machine needs to be transmitted to the correct applicatⁿ on another machine, addressing is provided by transport layer.

Q* Quality of Services & Transport service primitives:-

⇒ Quality of services refers to performance level of a service from perspective of the user.

In context of transport services, which encompasses various modes like transportation of goods or people. of data packets.

QOS:-

1) Reliability:-

The service should be dependable, ensuring that it operates consistently without interruptⁿ. Involves minimizing packet loss, error detection.

2) Safety/Security

Safety is paramount in transport services. includes Security measures to protect data integrity and confidentiality. Involves encryption, authentication, access control, & intrusion detection to safeguard network traffic.

3) Jitter:-

Inconsistent packet arrival times, known as jitter, can disrupt real-time communication. So, qos mechanisms mitigate jitter.

TSP:-

1) connectⁿ est^t :-

T-S primitives facilitate establishment, maintenance of connectⁿ betⁿ network endpoints.

2) Error control:-

TSP ensure error detection & correction during data transmission.

3) Flow control:-

regulate rate of data transmission betⁿ network endpoints to prevent congestion & buffer overflow.

4) Multiplex^g / Demultiplex^g:-

T-SP enables multiplexing of multiple data streams over a single network connection. & demultiplex^g at receiv^g end.

Components of Network.

1) NIC

Network Interface cards (NIC) are hardware components that used to connect computer to another comp. onto a network.

Two types:-

• Wired NIC:-

It is present in the motherboard. Cables & connectors are used with wired NIC to transfer data.

• Wireless NIC:-

It uses antenna to obtain the connection. For eg laptop & comp use Wireless NIC.

2) Hub:-

Hub is a hardware components which divides network connection with multiple devices.

When a computer request some info. it sends request to hub through cable. Hub broadcasts it to entire network. Then the devices checks whether the req belongs to them or not, If not, the req is dropped.

Hubs are not in that much use now, the more advanced comp. like switches & Routers are in existence.

3) Switch:-

hardware comp. which connects multiple devices to on a Network. They have more advanced features than hub. It just responsible for transmitting data to the correct destination based on physical address present in incoming message. It does not broadcast message to entire network.

4) Router:-

- A hardware device which is used to connect LAN to an Internet connection.
- It is used to receive, analyze, and forward incoming packets to another network.
- works in Layer 3 (Transport layer) of OSI model.

5) Cables & connectors:-

It is a transmission media used for transmitting a signal.

3 types of cables:-

1) Twisted-pair cables

2) Coaxial cables

3) Fibre-optic cables.

Short notes:-

- 1) Establishing a connectⁿ:-
- setting up communicatⁿ betⁿ two devices or nodes in network.
 - Includes series of steps, like handshake^g protocols, authenticatⁿ, etc.
 - Diff. network^g protocols have diff. methods for establishing connections.
 - For eg TCP uses three-way handshake to establish a connection, while UDP does not have explicit connectⁿ est^t phase.

2) Releasing a connectⁿ:-

- also known as connectⁿ termination.
- process of end^g communicatⁿ betⁿ two devices or nodes in network.
- In TCP, connectⁿ terminatⁿ follows four-way handshake process to ensure both parties agree to terminate.
- Properly releas^g connectⁿ is imp to free free up network resources.

3) Flow control.

Buffering:-

- Buffering is used in CN to temporarily store data packets when there is a difference in the rate of data flow betⁿ sender & receiver.
- are used to smooth out these diff. allowing the sender to continue transmitting data at its own pace while receiver processes data as it becomes available.
- implemented in d^g network devices such as switch, routers, etc.
- helps prevent data loss.

* Multiplexing:-

- combining multiple data streams into a single stream for transmission over a shared medium or communicⁿ channel.
- can be achieved using TDM, FDM, WDM (wavelength), etc.
- widely used in telecommunicatⁿ.

* Crash Recovery

- refers process of recovering from system failures or crashes to ensure the continued operation & integrity of network services.

- includes recovering from failure of network devices, such as routers or switches.
- Techniques involves fault tolerance mechanisms, such as backup syst, load balancing.
- essential for maintaining availability & reliability of network services in face of hardware or s/w failures.

IP Header:-

version	HLLEN	Type of service	Total length												
4 bits	4 bits	8 bits	16 bits												
Identificat ⁿ			<table border="1"> <tr> <td>Res</td> <td>DF</td> <td>MF</td> <td>Fragment</td> </tr> <tr> <td>2 bit</td> <td>1 bit</td> <td>1 bit</td> <td>offset</td> </tr> <tr> <td></td> <td></td> <td></td> <td>13 bits</td> </tr> </table>	Res	DF	MF	Fragment	2 bit	1 bit	1 bit	offset				13 bits
Res	DF	MF	Fragment												
2 bit	1 bit	1 bit	offset												
			13 bits												
16 bits															
Time to live	Protocol	Header checksum													
8 bits	8 bits	16 bits													
Source IP															
32 bits															
Dest IP															
32 bits															
Option															
0 to 40 Bytes															
Data															

Leaky Bucket

1) When a host wants to throw a packet packet is thrown in bucket.

2) Bucket leaks at constant rate.

3) used to regulate output rate

4) No Token generatⁿ

5) It does not save token

Token Bucket

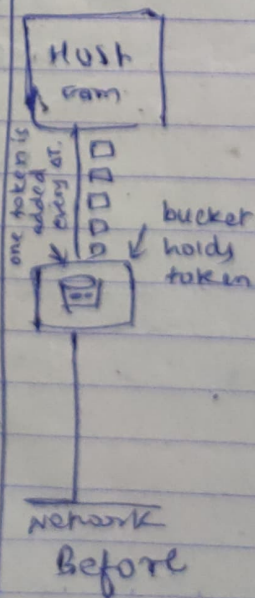
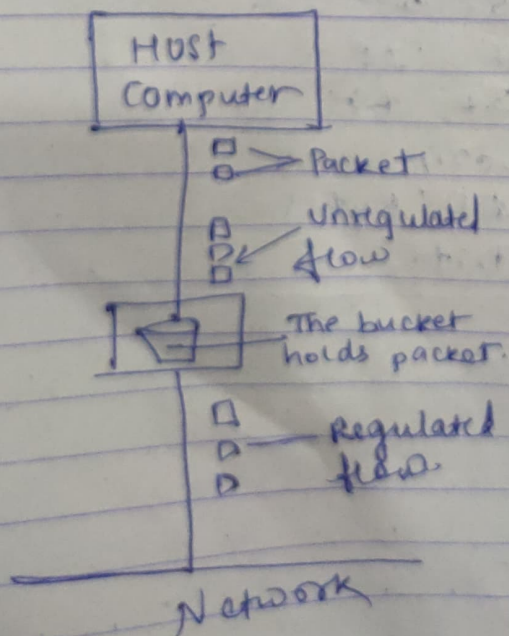
1) In this, bucket holds tokens generated at regular intervals of time.

2) Bucket has max. capacity.

3) used to regulate input rate.

4) Token is generated

5) It saves token to send large bursts.

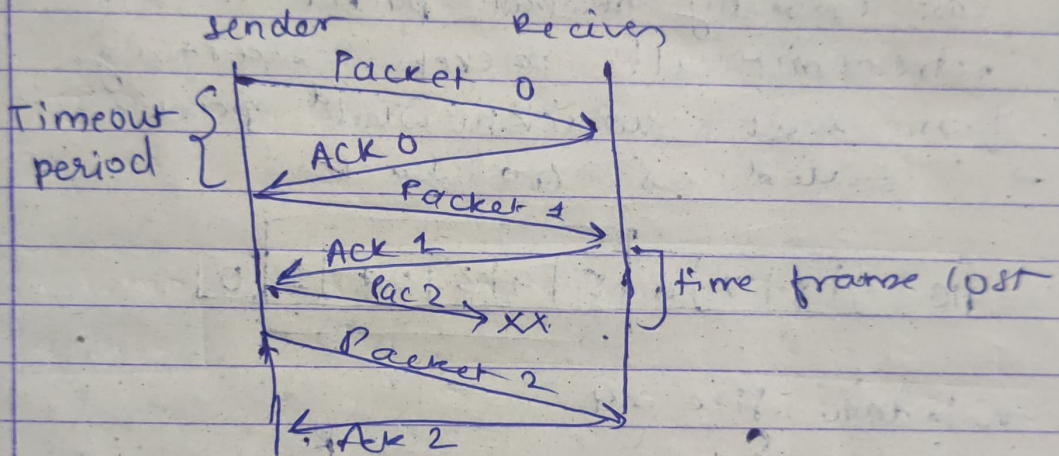


* Elementary Data Link Protocols:-

Noisy channels:-

1) Stop & Wait: ARQ:-

The sender sends the data and waits for a time of periods and wait for ACK before sending next frame.

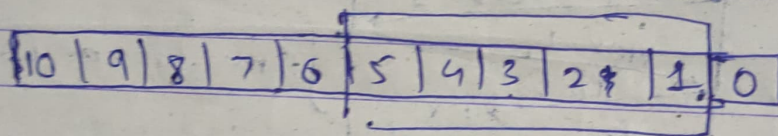


2) ~~Sliding window Protocol:-~~

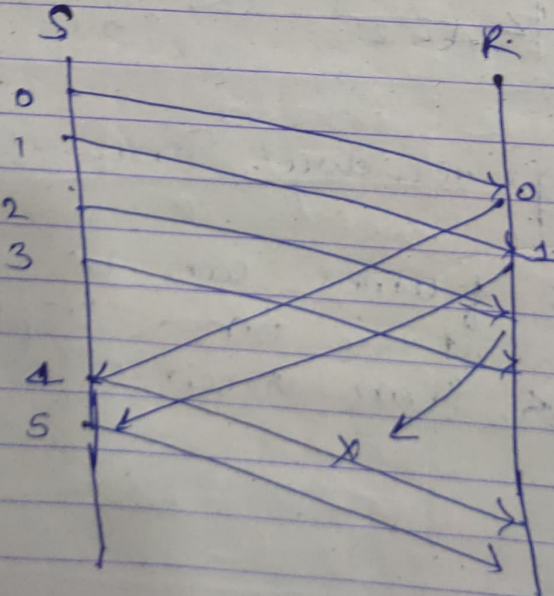
~~Multiple frames can be sent by a Sender at a time before receiving an ACK from receiver.~~

2) Go-Back-N 'ARG' :-

- sender is allowed to transmit multiple packets before waiting for an acknowledgment
- receiver receives data packets & sends ack only for last correctly received packet.
- If sender does not receive ACK for last correctly received packet it will retransmit all packets starting from last unacknowledged packet.
→ ∴ called as Go-back



Window size $\Rightarrow 4$.

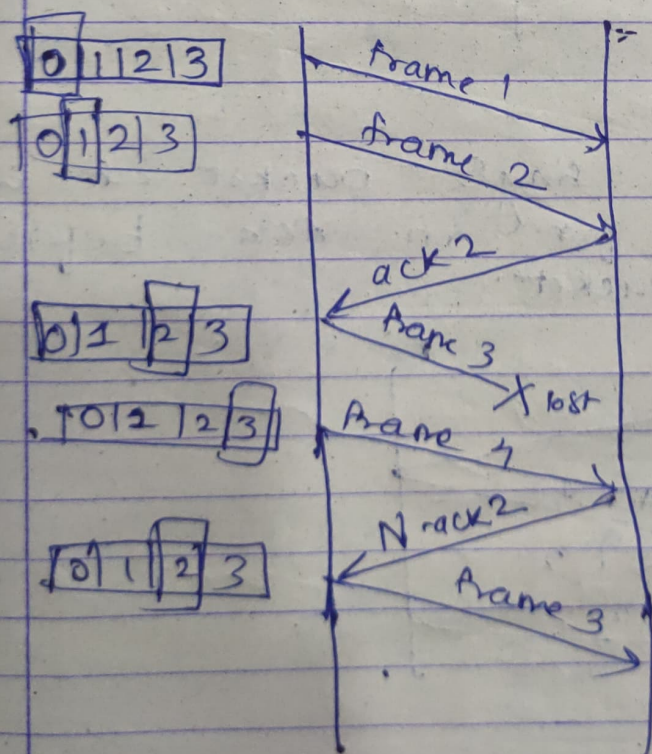


3) Repeat Selective ARQ

It is like Go back ARQ, which lets us to send multiple data packets before they receive ACK from server side.

But, there's small difference.

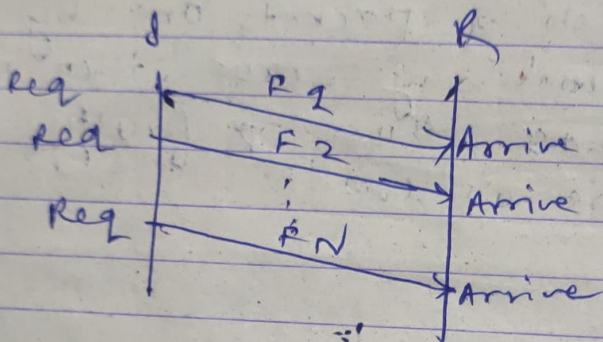
- the Selective ARQ does not request to retransmit entire window, as it checks each of packets one by one & if found any error, then only retransmits those faulty packages, not entire window.



Noiseless

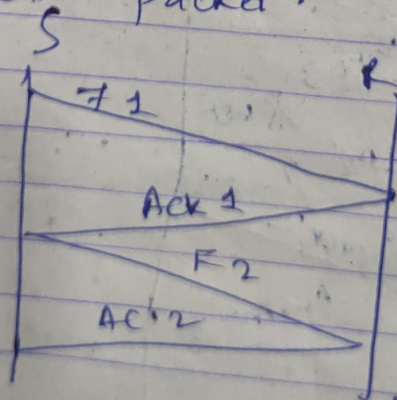
1) Simplest protocol:-

- involves direct transfer of data from source to destination without any intermediate processing
- assumption is that there will be no data loss



2) Stop-wait.

- Sender sends single packet at a time & wait for an Ack before sending next packet.



128 64 32 16 8 4 2 1
 1 0 0 1 1 0 1 1

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Hamming code

✓ 1011011 - even parity
 received code

7 bits

D ₇	D ₆	D ₅	P ₄	P ₃	P ₂	P ₁
1	0	1	1	0	1	1

$$P_1 = D_3 D_5 D_7$$

Put $P_1 = 1$

$$1 = 0 1 1 \rightarrow (1) \text{ error}$$

$$P_2 = D_3 D_6 D_7$$

$$1 = 0 0 1 \rightarrow (0) \text{ error}$$

$$P_4 \rightarrow D_5 D_6 D_7$$

Put $P_4 = 1$

$$1 \rightarrow 1 0 1 \rightarrow (1) \text{ error}$$

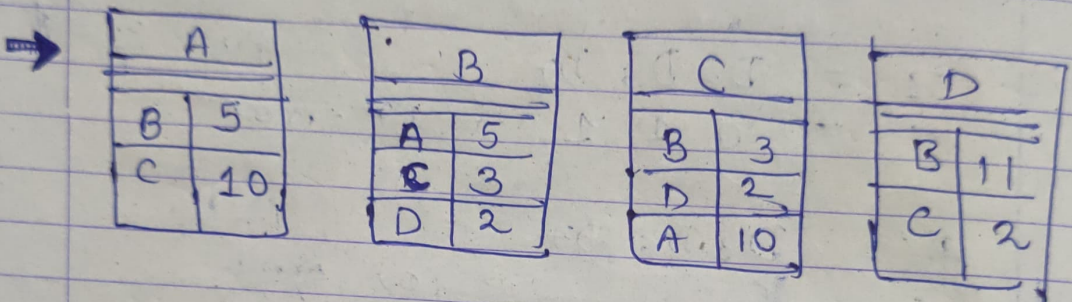
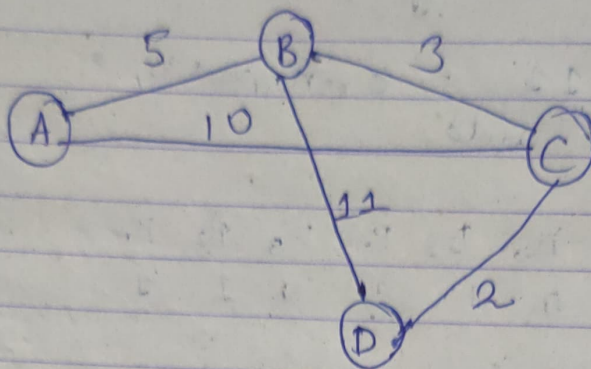
~~1010010~~
~~1001011~~
~~1010010~~

101 ← 5

(101), → 5 bits error

1001011

Link - State

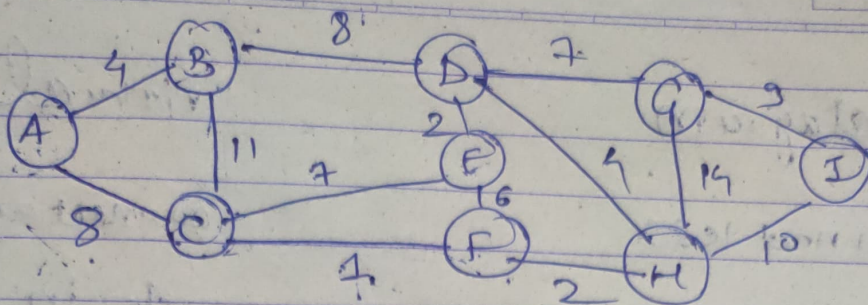


At A,

A	B	C	D
A	5	10	∞
AB	5	8	16
ABC	5	8	10
ABCD	5	8	10

	via
A	A
B	A
C	AB
D	ABC

shortest path is 10 From A to D.



→	A	B	C	D	E	F	G	H	I
A		(4)	8	∞	∞	∞	∞	∞	∞
AB		(4)	(8)	12	∞	∞	∞	∞	∞
ABC		(4)	(8)	12	15	(9)	∞	∞	∞
ABCF		(4)	(8)	12	15	(9)	∞	(11)	∞
ABCFH		(4)	(8)	(12)	14	(9)	28	(11)	21
ABCFHDE		(4)	(8)	(12)	(14)	(9)	19	(11)	21
ABCFHDGI		(4)	(8)	(12)	(14)	(9)	(19)	(11)	(21)

A	0
B	4 ✓
C	8 ✓
D	12 ✓
E	14 ✓
F	9 ✓
G	19 ✓
H	11 ✓
I	21 ✓

Datagram

Virtual

- | | | |
|---|----------------------------|-------------------------------|
| ① | can not less | connect oriented |
| ② | same sequence | each in order |
| ③ | not reliable | highly reliable |
| | cost efficient
easy | expensive |
| | widely used in
Internet | Asynchronous
Transfer mode |

10110011

 $x^7 + x + 1$

$$1 \cdot x^7 + 0 \cdot x^6 + 0 \cdot x^5 + 1 \cdot x^4 + 1 \cdot x^3 + 0 \cdot x^2 + 0 \cdot x + 1 \cdot x^0$$

10011

5 → 4 zero

	1010	
10011	101100110000	
	10011 ↓	
	001010	
	10011	
	11001	
	10011 ↓	
	010101	
	10011 ↓	
	001101	
	10011	
	011110	